

# 8

## Two Qubits, Three

*Not only is the Universe stranger than we think,  
it is stranger than we can think.*

Werner Heisenberg [2]

In the [previous chapter](#) we defined qubits and saw what we could do with just one of them. Things now start to get exponential with every additional qubit added, because entanglement allows the size of the working state space to double.

This chapter is about seeing how multiple qubits can behave together and then building a collection of tools to manipulate those qubits. These include the concept of entanglement, which is a requirement to do quantum computing. We also examine important 2-qubit gates like **CNOT** and **SWAP**. This will lead us into [chapter 9](#) and [chapter 10](#) where we look at algorithms and build circuits that use this machinery.

All vector spaces considered in this chapter are over  $\mathbb{C}$ , the field of complex numbers introduced in [section 3.9](#). All bases are orthonormal unless otherwise specified.

### Topics covered in this chapter

|       |                                         |     |
|-------|-----------------------------------------|-----|
| 8.1   | Tensor products . . . . .               | 270 |
| 8.2   | Entanglement . . . . .                  | 275 |
| 8.2.1 | Moving from one to two qubits . . . . . | 276 |

|       |                                                     |     |
|-------|-----------------------------------------------------|-----|
| 8.2.2 | The general case . . . . .                          | 281 |
| 8.2.3 | The density matrix again . . . . .                  | 283 |
| 8.3   | Multi-qubit gates . . . . .                         | 283 |
| 8.3.1 | The quantum $H^{\otimes n}$ gate . . . . .          | 284 |
| 8.3.2 | The quantum <b>SWAP</b> gate . . . . .              | 288 |
| 8.3.3 | The quantum <b>CNOT</b> / <b>CX</b> gate . . . . .  | 289 |
| 8.3.4 | The quantum <b>CY</b> and <b>CZ</b> gates . . . . . | 291 |
| 8.3.5 | The quantum $CR_\phi^Z$ gate . . . . .              | 293 |
| 8.3.6 | The quantum Toffoli <b>CCNOT</b> gate . . . . .     | 293 |
| 8.3.7 | The quantum Fredkin <b>CSWAP</b> gate . . . . .     | 294 |
| 8.4   | Summary . . . . .                                   | 295 |

## 8.1 Tensor products

In this section I introduce the linear algebra construction of a *tensor product*. If the direct sum seems to concatenate two vector spaces, then the tensor product interleaves them. In the first case, if we start with dimensions  $n$  and  $m$ , we end up with a new vector space of  $n + m$  dimensions. For the tensor product, we get  $nm$  dimensions.

We can quickly get vector spaces with high dimensions through this multiplicative effect. This means we need to use our algebraic intuition and tools more than our geometric ones.

The initial construction is straight linear algebra but we specialize it to quantum computing and working with multiple qubits in the next section.

Let  $V$  and  $W$  be two finite dimensional vector spaces over  $\mathbb{F}$ . Define a new vector space  $V \otimes W$ , pronounced “ $V$ ” tensor “ $W$ ” or “the tensor product of  $V$  and  $W$ ,” to be the vector space generated by addition and scalar multiplication of all objects  $\mathbf{v} \otimes \mathbf{w}$  for each  $\mathbf{v}$  in  $V$  and  $\mathbf{w}$  in  $W$ .

Note I said “generated by.” Not all vectors in  $V \otimes W$  look like  $\mathbf{v} \otimes \mathbf{w}$  for some  $\mathbf{v}$  in  $V$  and  $\mathbf{w}$  in  $W$ . A vector might look like  $2\mathbf{v}_1 \otimes \mathbf{w}_3 + 9\mathbf{v}_4 \otimes \mathbf{w}_7$ , for example.

Tensor products have the following properties:

- If  $a$  is a scalar in  $\mathbb{F}$  then

$$a(\mathbf{v} \otimes \mathbf{w}) = (a\mathbf{v}) \otimes \mathbf{w} = \mathbf{v} \otimes (a\mathbf{w})$$

## 8.1. Tensor products

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- For  $\mathbf{v}_1$  in  $V$  and  $\mathbf{w}_1$  and  $\mathbf{w}_2$  in  $W$ ,

$$\mathbf{v}_1 \otimes (\mathbf{w}_1 + \mathbf{w}_2) = \mathbf{v}_1 \otimes \mathbf{w}_1 + \mathbf{v}_1 \otimes \mathbf{w}_2$$

- For  $\mathbf{v}_1$  and  $\mathbf{v}_2$  in  $V$  and  $\mathbf{w}_1$  in  $W$ ,

$$(\mathbf{v}_1 + \mathbf{v}_2) \otimes \mathbf{w}_1 = \mathbf{v}_1 \otimes \mathbf{w}_1 + \mathbf{v}_2 \otimes \mathbf{w}_1$$

- If  $f : V \times W \rightarrow U$  is a bilinear map, then  $f^\otimes : V \otimes W \rightarrow U$  is a linear map defined by

$$f^\otimes (\mathbf{v} \otimes \mathbf{w}) = f(\mathbf{v}, \mathbf{w}).$$

- If  $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$  and  $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$  are bases for  $V$  and  $W$ , then

$$\begin{aligned} &\mathbf{v}_1 \otimes \mathbf{w}_1, \mathbf{v}_1 \otimes \mathbf{w}_2, \dots, \mathbf{v}_1 \otimes \mathbf{w}_m, \\ &\quad \vdots \\ &\mathbf{v}_n \otimes \mathbf{w}_1, \mathbf{v}_n \otimes \mathbf{w}_2, \dots, \mathbf{v}_n \otimes \mathbf{w}_m, \end{aligned}$$

are basis vectors of  $V \otimes W$ . There are  $nm$  of them and this is the dimension of  $V \otimes W$ .

For two additional vectors spaces  $X$  and  $Y$  over  $\mathbb{F}$ , if

$$f : V \rightarrow X \quad \text{and} \quad g : W \rightarrow Y$$

are linear maps, then so is

$$f \otimes g : V \otimes W \rightarrow X \otimes Y$$

where we define

$$(f \otimes g) (\mathbf{v} \otimes \mathbf{w}) = f(\mathbf{v}) \otimes g(\mathbf{w}).$$

### Question 8.1.1

Confirm  $f \otimes g$  is a linear map.

If  $f$  and  $g$  are both [monomorphisms](#), so is  $f \otimes g$ . If they are both [epimorphisms](#), so is  $f \otimes g$ .

If we have

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix}$$

then the regular matrix product is

$$AB = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} \\ a_{2,1}b_{1,1} + a_{2,2}b_{2,1} & a_{2,1}b_{1,2} + a_{2,2}b_{2,2} \end{bmatrix}$$

The matrix tensor product for 2 by 2 matrices is

$$\begin{aligned} A \otimes B &= \begin{bmatrix} a_{1,1}B & a_{1,2}B \\ a_{2,1}B & a_{2,2}B \end{bmatrix} \\ &= \begin{bmatrix} a_{1,1} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} & a_{1,2} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} \\ a_{2,1} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} & a_{2,2} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} a_{1,1}b_{1,1} & a_{1,1}b_{1,2} & a_{1,2}b_{1,1} & a_{1,2}b_{1,2} \\ a_{1,1}b_{2,1} & a_{1,1}b_{2,2} & a_{1,2}b_{2,1} & a_{1,2}b_{2,2} \\ a_{2,1}b_{1,1} & a_{2,1}b_{1,2} & a_{2,2}b_{1,1} & a_{2,2}b_{1,2} \\ a_{2,1}b_{2,1} & a_{2,1}b_{2,2} & a_{2,2}b_{2,1} & a_{2,2}b_{2,2} \end{bmatrix} \end{aligned}$$

and this is respect to the  $\mathbf{e}_1 \otimes \mathbf{e}_1$ ,  $\mathbf{e}_1 \otimes \mathbf{e}_2$ ,  $\mathbf{e}_2 \otimes \mathbf{e}_1$ , and  $\mathbf{e}_2 \otimes \mathbf{e}_1$  basis. I'm abusing the notation a bit in the second line to show we get a block of elements in the new matrix by multiplying an entry of the first by every entry in the second. The same kind of construction rules applies for larger matrices.

We do something similar with vectors. If  $\mathbf{v} = (v_1, v_2, v_3)$  in  $V$  and  $\mathbf{w} = (w_1, w_2)$  in  $W$ , then

$$\mathbf{v} \otimes \mathbf{w} = (v_1\mathbf{w}, v_2\mathbf{w}) = (v_1w_1, v_1w_2, v_2w_1, v_2w_2, v_3w_1, v_3w_2)$$

I've again taken liberties in the second term to show we get three entries by multiplying the second vector by an entry in the first. If  $V$  and  $W$  have [Euclidean norms](#), then

$$\begin{aligned} \|\mathbf{v} \otimes \mathbf{w}\|^2 &= \|(v_1w_1, v_1w_2, v_2w_1, v_2w_2, v_3w_1, v_3w_2)\|^2 \\ &= |v_1w_1|^2 + |v_1w_2|^2 + |v_2w_1|^2 + |v_2w_2|^2 + |v_3w_1|^2 + |v_3w_2|^2 \\ &= |v_1|^2|w_1|^2 + |v_1|^2|w_2|^2 + |v_2|^2|w_1|^2 + |v_2|^2|w_2|^2 + |v_3|^2|w_1|^2 + |v_3|^2|w_2|^2 \\ &= |v_1|^2(|w_1|^2 + |w_2|^2) + |v_2|^2(|w_1|^2 + |w_2|^2) + |v_3|^2(|w_1|^2 + |w_2|^2) \\ &= (|v_1|^2 + |v_2|^2 + |v_3|^2)(|w_1|^2 + |w_2|^2) \\ &= \|\mathbf{v}\|^2 \|\mathbf{w}\|^2 \end{aligned}$$

## 8.1. Tensor products

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This is true in general for finite dimensional vector spaces  $V$  and  $W$  with Euclidean norms.

For  $\mathbb{R}^2 \otimes \mathbb{R}^2$  or  $\mathbb{C}^2 \otimes \mathbb{C}^2$ ,

$$\mathbf{e}_1 \otimes \mathbf{e}_1 = (1, 0, 0, 0)$$

$$\mathbf{e}_1 \otimes \mathbf{e}_2 = (0, 1, 0, 0)$$

$$\mathbf{e}_2 \otimes \mathbf{e}_1 = (0, 0, 1, 0)$$

$$\mathbf{e}_2 \otimes \mathbf{e}_2 = (0, 0, 0, 1)$$

Tensor products combine nicely with traditional products. For two additional matrices,

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

We now take a brief culinary diversion to again compare direct sums and tensor products. Let  $V$  be the vector space with basis **chocolate ice cream**, **vanilla ice cream**, and **mint chocolate chip ice cream**. For the second vector space  $W$ , the basis is **chocolate fudge sauce**, **caramel sauce**, **mango sauce**, and **raspberry sauce**.

Vectors in  $V \oplus W$  are linear combinations of the 7 ( $= 3 + 4$ ) foods

**chocolate ice cream**  
**vanilla ice cream**  
**mint chocolate chip ice cream**  
**chocolate fudge sauce**  
**caramel sauce**  
**mango sauce**  
**raspberry sauce**

Vectors in  $V \otimes W$  are linear combinations of the 12 ( $= 3 \times 4$ ) food combinations:

**chocolate ice cream  $\otimes$  chocolate fudge sauce**  
**chocolate ice cream  $\otimes$  caramel sauce**  
**chocolate ice cream  $\otimes$  mango sauce**  
**chocolate ice cream  $\otimes$  raspberry sauce**  
  
**vanilla ice cream  $\otimes$  chocolate fudge sauce**  
**vanilla ice cream  $\otimes$  caramel sauce**  
**vanilla ice cream  $\otimes$  mango sauce**

vanilla ice cream  $\otimes$  raspberry sauce

mint chocolate chip ice cream  $\otimes$  chocolate fudge sauce

mint chocolate chip ice cream  $\otimes$  caramel sauce

mint chocolate chip ice cream  $\otimes$  mango sauce

mint chocolate chip ice cream  $\otimes$  raspberry sauce

We get every possible pairing with the tensor product. I think the last three are the most questionable flavor mixtures. Now back to the math.

If  $A$  and  $B$  are **unitary matrices** then so is  $A \otimes B$ .

### Question 8.1.2

For the two unitary matrices

$$H = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \quad \text{and} \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

which are the Hadamard and Pauli  $\sigma_y$  matrices, respectively, show

$$H \otimes \sigma_y = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{2}i & 0 & -\frac{\sqrt{2}}{2}i \\ \frac{\sqrt{2}}{2}i & 0 & \frac{\sqrt{2}}{2}i & 0 \\ 0 & -\frac{\sqrt{2}}{2}i & 0 & \frac{\sqrt{2}}{2}i \\ \frac{\sqrt{2}}{2}i & 0 & -\frac{\sqrt{2}}{2}i & 0 \end{bmatrix}$$

and it is unitary.

Consider  $\mathbb{C}^2$ . Both  $\mathbb{C}^2 \oplus \mathbb{C}^2$  and  $\mathbb{C}^2 \otimes \mathbb{C}^2$  have four dimensions. These are *isomorphic*: we have an invertible linear map from all of the first vector space to all of the second. What is it?

Let  $\mathbf{e}_1 = (1, 0)$  and  $\mathbf{e}_2 = (0, 1)$  be the standard basis on  $\mathbb{C}^2$ . Since  $\mathbb{C}^2 \oplus \mathbb{C}^2 = \mathbb{C}^4$ , it has the standard basis  $\mathbf{f}_1 = (1, 0, 0, 0)$ ,  $\mathbf{f}_2 = (0, 1, 0, 0)$ ,  $\mathbf{f}_3 = (0, 0, 1, 0)$ , and  $\mathbf{f}_4 = (0, 0, 0, 1)$ . (I'm using “**f**” instead of “**e**” in this second case to avoid confusion.)

## 8.2. Entanglement

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Define  $S : \mathbb{C}^2 \oplus \mathbb{C}^2 = \mathbb{C}^4 \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^2$  by

$$\begin{array}{ll} \mathbf{f}_1 \mapsto \mathbf{e}_1 \otimes \mathbf{e}_1 & \mathbf{f}_3 \mapsto \mathbf{e}_2 \otimes \mathbf{e}_1 \\ \mathbf{f}_2 \mapsto \mathbf{e}_1 \otimes \mathbf{e}_2 & \mathbf{f}_4 \mapsto \mathbf{e}_2 \otimes \mathbf{e}_2 \end{array}$$

This is not the only isomorphism but it is natural given our calculation of the coordinates of the vectors like  $\mathbf{e}_1 \otimes \mathbf{e}_2$  above.

Another interesting basis for  $\mathbb{C}^2 \otimes \mathbb{C}^2$  is

$$\begin{array}{ll} \frac{\sqrt{2}}{2} (\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2) & \frac{\sqrt{2}}{2} (\mathbf{e}_1 \otimes \mathbf{e}_1 - \mathbf{e}_2 \otimes \mathbf{e}_2) \\ \frac{\sqrt{2}}{2} (\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1) & \frac{\sqrt{2}}{2} (\mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_2 \otimes \mathbf{e}_1) \end{array}$$

Continuing on,  $\mathbb{C}^2 \oplus \mathbb{C}^2 \oplus \mathbb{C}^2 = \mathbb{C}^6$  but  $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$  has 8 dimensions. If we tensor ten copies of  $\mathbb{C}^2$  together we get  $2^{10} = 1024$  dimensions.

The process of tensoring with more copies of  $\mathbb{C}^2$  is exponential in the number of dimensions.

### To learn more

Tensor products are not always included in introductory texts on linear algebra, and so you may not have seen them before even if you have studied the subject. More complete mathematical treatments cover tensor products of vectors and matrices, as I have, but may also generalize them using category theory. [1] [3] [4, Monoids]

In the next section, we use the tensor product as the underpinning of qubit entanglement and show what they look like and how they behave with bra-ket notation.

## 8.2 Entanglement

We've now seen many gate operations that you can apply to a single qubit to change its state. In [section 2.5](#) we worked through how to apply classical logic gates to build a circuit for addition.

While we can apply **not** to a single bit, all the other operations require at least two bits for input. In the same way, we need to work with multiple qubits to produce interesting and useful results.